PERMUTATION & COMBINATION

- **Factorial Notation**: The product of first n natural numbers is called as factorial n and is denoted by $n! = \underline{|n} = 1 \times 2 \times 3 \times 4 \times ... \times n$.
  (1) We have to assume $0! = 1$, though it makes no sense.
  (2) When n is a negative integer then $n! = \underline{|n}$ is not defined.
  (3) $n! = \underline{|n} = 1 \times 2 \times 3 \times 4 \times ... \times n = n \times 1 \times 2 \times 3 \times 4 \times ... \times (n-1)$
  $\Rightarrow n! = n \times (n-1)!$.
  (4) $(2n)! = 2^n \times n! \times (1 \times 3 \times 5 \times ... \times (2n-1))$.

- **Multiplication Principle**: If an operation can be performed in $'m'$ different ways and second operation can be performed in $'n'$ different ways, then the two operations in succession can be performed in $m \times n$ different ways.

- **Addition Principle**: If an operation can be performed in $'m'$ different ways and second operation which is independent of first operation, can be performed in $'n'$ different ways, then the two operations can be performed in $m + n$ different ways.

- **Permutation**: Permutation means arrangement. Each of the different arrangements which can be made by taking some or all of given number of things or objects at a time is called a permutation.

- **Combination**: It means selection, Chose, or making committee. Each of the different groups or selections which can be made by taking some or all of a number of things (irrespective of order) is called as combination.

- ${}^nP_r = P(n,r) = \dfrac{n!}{(n-r)!} = n(n-1)(n-2)(n-3)...(n-r+1),\ 0 \le r \le n$.
- ${}^nP_r$ is always a positive integer.
- $\dfrac{{}^nP_r}{{}^nP_{-1r}} = n - r + 1$.
- ${}^nP_r = n.{}^{n-1}P_{r-1} = n(n-1)^{n-2}P_{r-2} = n(n-1)(n-2)^{n-3}P_{r-3} = ...$
- ${}^nP_r + r\,{}^nP_{r-1} = {}^{n+1}P_r$.
- ${}^nP_r = {}^{n-1}P_r + r\,{}^{n-1}P_{r-1}$.
- Number of permutations of $'n'$ different things taken all at a time is ${}^nP_n = n!$.
- The number of permutations of $'n'$ things taken all at a time, out of which $'p'$ are alike and are of one type and $'q'$ are alike and are of second type and rest all are different is $\dfrac{n!}{p!.q!}$.
- The number of permutation of $'n'$ different things taken $'r'$ at a time when each thing may be repeated any number of times is n^r.

- The number of permutation of 'n' different things taken 'r' at a time, when a particular things is to be always included in each arrangement, is $r.^{n-1}P_{r-1}$.
- The number of permutations of 'n' different things taken 'r' at a time, when 's' particular things are to be always included in each arrangement is ${}^{r}P_{s}.^{n-1}P_{r-s} = r! \times {}^{n-s}C_{r-s}$.
- The number of permutation of 'n' different things taken 'r' at a time, when a particular things is never taken in each time is ${}^{n-1}P_r$.
- The number of permutation of 'n' different things taken all at a time, when 'm' specified things always come together is $m! \times (n-m+1)!$.
- The number of permutation of 'n' different things taken all at a time, when 'm' specified things never come together is $n! - (m! \times (n-m+1)!)$.
- The number of permutation of 'n' different things taken 'r' at a time, when atleast one thing is repeated is $n^r - {}^nP_r$.
- The number of permutation of 'n' different things taken 'r' at a time, when each thing may occur any number of times is $\frac{n(n^r - 1)}{n-1}$.
- Let there are 'n' objects, of which 'm' objects are alike of one kind, and the remaining '$n-m$' objects are alike of another kind. Then the total number of distinct permutations that can be formed from these objects is $\frac{n!}{m! \times (n-m)!}$.
- Let there are 'n' objects, of which 'p_1' objects are alike of one kind, 'p_2' are alike of 2nd kind,…, 'p_k' are alike of kth kind such that $p_1 + p_2 + ... + p_k = n$. Then the total number of distinct permutations that can be formed from these objects is $\frac{n!}{(p_1!) \times (p_2!) \times ... \times (p_k!)}$.
- The number of significant numbers consisting of 'r' digits and formed out of 'n' digits including zero (0), no digit being repeated in any number is ${}^nP_r - {}^{n-1}P_{r-1}$.
- The number of permutations of 'n' different things where order of 'r' things is not to be considered is $\frac{n!}{r!}$.
- The number of circular permutations of 'n' different things is $(n-1)!$.
- The number of circular arrangements of 'n' different things when clock wise and anti-clock wise arrangements are not different is $\frac{(n-1)!}{2}$.
- The number of circular arrangements of 'n' different things taken when 'r' at a time, when clock wise and anti-clock orders are taken as different is $\frac{{}^nP_r}{r}$.
- The number of circular arrangements of 'n' different things taken when 'r' at a time, when clock wise and anti-clock orders are not different is $\frac{{}^nP_r}{2r}$.

- **Derangement (or) Matching Problem**: If $'n'$ things form an arrangement in a row, the number of ways in which they can be arranged so that no one of them occupies its original place is $n!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-...+(-1)^n\frac{1}{n!}\right)$.
- The sum of digits in unit places of all numbers formed with the help of $a_1, a_2, ..., a_n$ all at a time is $(n-1)!(a_1+a_2+...+a_n)$.
- The sum of all possible numbers formed out of all the $'n'$ digits without zero is $=(n-1)!(\text{sum of all the digits})\times(111...\text{n times})$.
  $=(n-1)!(\text{sum of all the digits})\times\left(\frac{10^n-1}{9}\right)$.
- The sum of all possible numbers formed out of all the $'n'$ digits which includes zero is $=(\text{sum of all the digits})\left[(n-1)!(111...\text{n times})\text{ - }(n-2)!(111...(\text{n - 1})\text{ times})\right]$.
  $=(\text{sum of all the digits})\left[(n-1)!\left(\frac{10^n-1}{9}\right)\text{- }(n-2)!\left(\frac{10^{n-1}-1}{9}\right)\right]$
- Sum of all $'r'$ digits number formed by taking the given $'n'$ digits (including 0) is $=(\text{sum of all the digits})[{}^{n-1}P_{r-1}\times(111...\text{r times})\text{ - }{}^{n-2}P_{r-2}\times(111...(\text{r - 1})\text{ times})]$.
- Sum of all $'r'$ digits number formed by taking the given $'n'$ digits (excluding 0) is $=(\text{sum of all the digits})\times{}^{n-1}P_{r-1}\times(111...\text{r times})$.
- The number of mappings (functions) that can be defined from a set A containing $'m'$ elements to set B containing $'n'$ elements is n^m.
- The number of one-one (injective) mappings (functions) that can be defined from a set A containing $'m'$ elements to set B containing $'n'$ elements is ${}^nP_m$ where $m \le n$.
- The number of one-one onto (bijective) mappings (functions) that can be defined from a set A containing $'n'$ elements to set B containing $'n'$ elements is ${}^nP_n = n!$.
- The number of onto (surjective) mappings (functions) that can be defined from a set A containing $'m'$ elements to set B containing $'n'$ elements or the number of $'n'$ different things can be distributed into $'r'$ different groups where each group must have atleast one thing is $n^m - {}^nC_1(n-1)^m + {}^nC_2(n-2)^m - {}^nC_3(n-3)^m + ...$ where ${}^nC_r = 0 \;\; for \;\; n<r$.
- ${}^nC_r = \binom{n}{r} = C(n,r) = \frac{n!}{(n-r)!.r!} = \frac{{}^nP_r}{r!}$.
- ${}^nC_r = \frac{n(n-1)(n-2)...(n-r+1)}{r(r-1)(r-2)...1}$.
- ${}^nC_0 = {}^nC_n = 1$.
- ${}^nC_{n-1} + {}^nC_1 = n$.
- ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$.
- ${}^nC_r = {}^nC_{n-r}$.
- $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$.

- $\frac{^{n}C_{r}}{^{n-}C_{r-1}}=\frac{n}{r}$.
- $\frac{^{n}C_{r-1}}{^{n-1}C_{r-1}}=\frac{n}{n-r+1}$.
- If $^{n}C_{r}={}^{n}C_{s}$ then either $r=s$ or $r+s=n$.
- If $^{n}C_{r-1},{}^{n}C_{r},{}^{n}C_{r+1}$ are in A.P. then $(n-2r)^{2}=(n+2)$.
- If $'n'$ is even, then greatest value of $^{n}C_{r}$ is $={}^{n}C_{n/2}$.
- If $'n'$ is odd, then greatest value of $^{n}C_{r}$ is $={}^{n}C_{(n-1)/2}$ or $={}^{n}C_{(n+1)/2}$.
- $^{n}C_{0}+{}^{n}C_{1}+{}^{n}C_{2}+...+{}^{n}C_{n}=2^{n}$.
- $^{n}C_{0}+{}^{n}C_{2}+{}^{n}C_{4}+...={}^{n}C_{1}+{}^{n}C_{3}+{}^{n}C_{5}+...=2^{n-1}$.
- The number of combination of $'n'$ different things taken $'r'$ at a time, when any object may be repeated any number of times:
  $=Coefficient\ of\ \ x^{r}\ \ in\ \left(1+x+x^{2}+x^{3}+...+x^{r}\right)^{n}=Coefficient\ of\ \ x^{r}\ \ in\ (1-x)^{-n}={}^{n+r-1}C_{r}$
- The total number of ways in which it is possible to make groups by taking some or all out of $n=n_{1}+n_{2}+n_{3}+...+n_{r}$ things when n_{1} are alike of one kind, n_{2} are alike of 2nd kind and n_{r} are alike of rth kind is $=(n_{1}+1)(n_{2}+1)(n_{3}+1)...(n_{r}+1)-1$.
- The number of combination of $'n'$ different things taken $'r'$ at a time, when $'p'$ particular things is to be always included, is $={}^{n-p}C_{r-p}$.
- The number of combination of $'n'$ different things taken $'r'$ at a time, when $'p'$ particular things are never included, is $={}^{n-p}C_{r}$.
- The number of combination of $'n'$ different things taken $'r'$ at a time, when $'p'$ particular things are not together in any selection, is $={}^{n}C_{r}-{}^{n-p}C_{r-p}$.
- The number of selections of $'r'$ consecutive things out of $'n'$ things in a row is $=n-r+1$.
- The number of selections of $'r'$ consecutive things out of $'n'$ things along a circle is $=n \quad if\,(r<n)$.
- The number of selections of $'r'$ consecutive things out of $'n'$ things along a circle is $=1 \quad if\,(r=n)$.
- The number of combinations of $'n'$ different things taken any number (0 or more) at a time is $=2^{n}$ i.e. $^{n}C_{0}+{}^{n}C_{1}+{}^{n}C_{2}+...+{}^{n}C_{n}=2^{n}$.
- The number of combinations of $'n'$ different things taken 1 or more at a time is $={}^{n}C_{1}+{}^{n}C_{2}+...+{}^{n}C_{n}=2^{n}-1$.
- The number of selection of $'r'$ things out of $'n'$ identical things is = 1.
- The number of selection of 0 or more things out of $'n'$ identical things is $=n+1$.
- The number of selection of 1 or more things out of $'n'$ identical things is $=n$.
- The number of combinations of $'n'$ objects of which $'p'$ are identical , taken $'r'$ at a time is
  $$=\left[\begin{array}{ll} ^{n-p}C_{r}+{}^{n-p}C_{r-1}+{}^{n-p}C_{r-2}+...+{}^{n-p}C_{0} & r\le p \\ ^{n-p}C_{r}+{}^{n-p}C_{r-1}+{}^{n-p}C_{r-2}+...+{}^{n-p}C_{r-p} & r>p \end{array}\right.$$

- The number of ways in which $'n'$ different things can be arranged into $'r'$ different groups is $= {}^{n+r-1}P_n = n! \times {}^{n-1}C_{r-1}$ according as blanks group are not admissible.
- The number of ways in which $'n'$ different things can be distributed into $'r'$ different groups is $= r^n - {}^rC_1(r-1)^n + {}^rC_2(r-2)^n - ... + (-1)^{r-1}\,{}^rC_{r-1}$ according as blanks group are not admissible.
- The number distribution of $'n'$ different things into $'r'$ different groups in which no one of $'x'$ of the $'x'$ assigned parcels are blanks is $= r^n - {}^rC_1(r-1)^n + {}^rC_2(r-2)^n - ... + (-1)^{r-1}\,{}^rC_{r-1}$.
- The number of ways in which $m+n$ different things can be divided equally into $'n'$ groups is $= \dfrac{(m \times n)!n!}{(m!)^n.n!} = \dfrac{(m \times n)!}{(m!)^n}$.
- The number of ways in which $m+n$ different things can be divided into two different groups of $'m'$ and $'n'$ things separately is $= {}^{m+n}C_m\,{}^nC_n = \dfrac{(m+n)!}{m!.n!}$ where $m \neq n$.
- The number of ways in which $m+n+p$ different things can be divided into three different groups of $'m'$, $'n'$ and $'p'$ things separately is $= \dfrac{(m+n+p)!}{m!.n!.p!}$ where $m \neq n \neq p$.
- The number of ways of dividing $'2n'$ different things into two groups each containing $'n'$ things and order of the groups is not important is $= \dfrac{(2n)!}{2!.(n!)^2}$.
- The number of ways of dividing $'2n'$ different things into two groups each containing $'n'$ things and order of the groups is important is $= \dfrac{(2n)!}{(n!)^2}$.
- The number of ways of dividing $'3n'$ different things into three groups each containing $'n'$ things and order of the groups is not important is $= \dfrac{(3n)!}{3!.(n!)^3}$.
- The number of ways of dividing $'3n'$ different things into three groups each containing $'n'$ things and order of the groups is important is $= \dfrac{(3n)!}{(n!)^3}$.
- The number of ways of dividing $'kn'$ different things into $'k'$ groups each containing $'n'$ things and order of the groups is not important is $= \dfrac{(kn)!}{k!.(n!)^k}$.
- The number of ways of dividing $'kn'$ different things into $'k'$ groups each containing $'n'$ things and order of the groups is important is $= \dfrac{(kn)!k!}{(n!)^k k!} = \dfrac{(kn)!}{(n!)^k}$.
- If out of $p+q+r+t$ things $'p'$ are alike of one kind, $'q'$ are alike of second kind, $'r'$ are alike of third kind and $'t'$ are different, then the total number of selection is $= (p+1)(q+1)(r+1)2^t - 1$.
- If out of $p+q+r$ things $'p'$ are alike of one kind, $'q'$ are alike of second kind, $'r'$ are alike of third kind, then the total number of selection is $= (p+1)(q+1)(r+1) - 1$.

- Let $n = a_1^p.a_2^q.a_3^r... \neq 1$ where $a_1, a_2, a_3...$ are primes and $p, q, r \in N$, then number of all positive integers which are less than $'n'$ and are prime to it is $= N\left(1-\frac{1}{a_1}\right)\left(1-\frac{1}{a_2}\right)\left(1-\frac{1}{a_3}\right)...$
- Let $n = a_1^p.a_2^q.a_3^r... \neq 1$ where $a_1, a_2, a_3...$ are primes and $p, q, r \in N$, then number of divisors of $'n'$ is $= (p+1)(q+1)(r+1)....$
- Let $n = a_1^p.a_2^q.a_3^r... \neq 1$ where $a_1, a_2, a_3...$ are primes and $p, q, r \in N$, then number of divisors of $'n'$ excluding $'1'$ is $= (p+1)(q+1)(r+1)...-1$.
- Let $n = a_1^p.a_2^q.a_3^r... \neq 1$ where $a_1, a_2, a_3...$ are primes and $p, q, r \in N$, then number of divisors of $'n'$ excluding $'n'$ is $= (p+1)(q+1)(r+1)...-1$.
- Let $n = a_1^p.a_2^q.a_3^r... \neq 1$ where $a_1, a_2, a_3...$ are primes and $p, q, r \in N$, then number of proper divisors (non trivial divisors) of $'n'$ is $= (p+1)(q+1)(r+1)...-2$.
- Let $n = a_1^p.a_2^q.a_3^r... \neq 1$ where $a_1, a_2, a_3...$ are primes and $p, q, r \in N$, then number of improper divisors (trivial divisors) of $'n'$ is $= 2$.
- Let $n = a_1^{p_1}.a_2^{p_2}.a_3^{p_3}...a_k^{p_k} \neq 1$ is a positive integer where $a_1, a_2, a_3..., a_k$ are primes and $p_1, p_2, p_3..., p_k \in N$ then the number of divisors of $'n'$ is $= (p_1+1)(p_2+1)(p_3+1)...(p_k+1)$.
- Let $n = a_1^{p_1}.a_2^{p_2}.a_3^{p_3}...a_k^{p_k} \neq 1$ is a positive integer where $a_1, a_2, a_3..., a_k$ are primes and $p_1, p_2, p_3..., p_k \in N$ then the sum of divisors of $'n'$ is $= \frac{a_1^{(p_1+1)}-1}{a_1-1} \times \frac{a_2^{(p_2+1)}-1}{a_2-1} \times \frac{a_3^{(p_3+1)}-1}{a_3-1} \times ... \times \frac{a_k^{(p_k+1)}-1}{a_k-1}$.
- Let $n = a_1^{p_1}.a_2^{p_2}.a_3^{p_3}...a_k^{p_k}$ where $a_1, a_2, a_3..., a_k$ are primes other than 2 then number of even divisors is $= p_1(p_2+1)(p_3+1)...(p_k+1)$.
- Let $n = a_1^{p_1}.a_2^{p_2}.a_3^{p_3}...a_k^{p_k}$ where $a_1, a_2, a_3..., a_k$ are primes other than 2 then number of odd divisors is $= (p_2+1)(p_3+1)...(p_k+1)$.
- Let $n = a_1^{p_1}.a_2^{p_2}.a_3^{p_3}...a_k^{p_k}$ where $a_1, a_2, a_3..., a_k$ are primes other than 2 then number of ways in which $'n'$ can be resolved into 2 factors if $'n'$ is not a perfect square is $= \frac{1}{2}(p_1+1)(p_2+1)(p_3+1)...(p_k+1)$.
- Let $n = a_1^{p_1}.a_2^{p_2}.a_3^{p_3}...a_k^{p_k}$ where $a_1, a_2, a_3..., a_k$ are primes other than 2 then number of ways in which $'n'$ can be resolved into 2 factors if $'n'$ is a perfect square is $= \frac{1}{2}\{(p_1+1)(p_2+1)(p_3+1)...(p_k+1)+1\}$.
- The number of ways in which a composite number N can be resolved into two factors which are relatively prime (or co-prime) to each other is equal to 2^{n-1} where n is the number of different factors in N.
- The number of lines formed out of $'n'$ points in a plane of which no three points are collinear is $= {}^nC_2$.

- The number of lines formed out of $'n'$ points in a plane of which $'m'$ points are collinear is $={}^nC_2-{}^mC_2+1$.
- The number of triangles formed out of $'n'$ points in a plane of which $'m'$ points are collinear is ${}^nC_3-{}^mC_3$.
- The number of diagonals of a polygon of $'n'$ sides is ${}^nC_2-n=\frac{n(n-3)}{2}$.
- The number of parallelograms that can be formed from a set of $'n'$ parallel lines intersecting another set of $'m'$ parallel lines is ${}^nC_2\times{}^mC_2$.
- If $'n'$ distinct points are given on the circumference of a circle, then:
  (1) No. of Straight lines $={}^nC_2$.
  (2) No. of Trinagles $={}^nC_3$.
  (3) No. of Quadrilaterals $={}^nC_4$.
- The number of rectangles of any size in a square of $n\times n$ is $\sum_{r=1}^{n}r^3$.
- The number of squares of any size in a square of $n\times n$ is $\sum_{r=1}^{n}r^2$.
- The number of rectangles of any size of $n\times p$ where $n<p$ is $={}^nC_2\times{}^pC_2=\frac{np(n+1)(p+1)}{4}$.
- The number of squares of any size of $n\times p$ where $n<p$ is $\sum_{r=1}^{n}(n+1-r)(p+1-r)$.
- The number of ways in which $'n'$ identical things can be distributed into $'r'$ different groups is ${}^{n+r-1}C_{r-1}$.
- The number of ways in which $'n'$ identical things can be distributed into $'r'$ different groups so that each group to receive at-least one things is ${}^{n-1}C_{r-1}$.
- The number of non-negative integral solutions of $x_1+x_2+x_3+...+x_r=n$ is ${}^{n+r-1}C_{r-1}$.
- The number of positive integral solutions of $x_1+x_2+x_3+...+x_r=n$ is ${}^{n-1}C_{r-1}$.
- The number of ways of selection of $'r'$ things from $'n'$ different groups when each group may have identical things of $'r'$ or more things is ${}^{n+r-1}C_r$.
- **Exponent of Prime *p* in *n!*** : Exponent of p in n! is denoted by $E_p(n!)=\left[\frac{n}{p}\right]+\left[\frac{n}{p^2}\right]+\left[\frac{n}{p^3}\right]+...+\left[\frac{n}{p^k}\right]$ where $p^k<n<p^{k+1}$ and $\left[\frac{n}{p}\right]$ denotes the greatest integer function.

  Example: Exponent of 3 in 100! = $E_3(100!)=\left[\frac{100}{3}\right]+\left[\frac{100}{3^2}\right]+\left[\frac{100}{3^3}\right]+\left[\frac{100}{3^4}\right]$

  $$\Rightarrow E_3(100!)=\left[\frac{100}{3}\right]+\left[\frac{100}{9}\right]+\left[\frac{100}{27}\right]+\left[\frac{100}{81}\right]$$

  $$\Rightarrow E_3(100!)=33+11+3+1=48 \text{ (ANS)}$$
- $(n!)!$ is divisible by $(n!)^{(n-1)!}$.

- $(kn)!$ is divisible by $(n!)^k$.
- ${}^nC_r$ is divisible by n if n is a prime and $n \neq r$.

- **Multinomial Theorem**: Let $x_1, x_2, x_3, ..., x_m$ be integers. Then number of solution to the equation $x_1 + x_2 + x_3 + ... + x_m = n$ subject to the condition $a_1 \le x_1 \le b_1, a_2 \le x_2 \le b_2, ..., a_m \le x_m \le b_m$ is equal to the coefficient of x^n in the expansion of: $(x^{a_1} + x^{a_1+1} + ... + x^{b_1})(x^{a_2} + x^{a_2+1} + ... + x^{b_2})...(x^{a_m} + x^{a_m+1} + ... + x^{b_m})$.

- The number of integral solution of $x_1 + x_2 + x_3 + ... + x_m = n$ where $x_1 \ge 0, x_2 \ge 0, ..., x_m \ge 0$ is same as the number of ways to distribute n identical things among m persons

  $=$ the coefficient of x^n in the expansion of: $(1 + x + x^2 + x^3 + ...)^m$

  $=$ coefficient of x^n in $\left(\dfrac{1}{1-x}\right)^m$

  $=$ coefficient of x^n in $(1-x)^{-m}$

  $=$ coefficient of x^n in $\left(1 + {}^mC_1 x + {}^{m+1}C_2 x^2 + {}^{m+2}C_3 x^3 + ...\right)$

  $= {}^{m+n-1}C_n$

  $= {}^{m+n-1}C_{m-1}$.

- The number of natural solution of $x_1 + x_2 + x_3 + ... + x_m = n$

  $=$ the coefficient of x^n in the expansion of: $(x + x^2 + x^3 + ...)^m$

  $=$ the coefficient of x^n in the expansion of: $x^m(1 + x + x^2 + x^3 + ...)^m$

  $=$ the coefficient of x^{n-m} in the expansion of: $(1 + x + x^2 + x^3 + ...)^m$

  $=$ coefficient of x^{n-m} in $\left(\dfrac{1}{1-x}\right)^m$

  $=$ coefficient of x^{n-m} in $(1-x)^{-m}$

  $=$ coefficient of x^{n-m} in $\left(1 + {}^mC_1 x + {}^{m+1}C_2 x^2 + {}^{m+2}C_3 x^3 + ...\right)$

  $= {}^{m+n-m-1}C_{n-m}$

  $= {}^{n-1}C_{m-1}$

- The number of possible arrangement of n objects out of of which $'p_1'$ objects are alike of one kind, $'p_2'$ are alike of 2nd kind,…, $'p_k'$ are alike of kth kind such that $p_1 + p_2 + ... + p_k = n$ is:

  $= (n!) \times$Coefficient of x^n in the expansion of $\times$

  $$\left(1 + x + \frac{x^2}{2!} + ... + \frac{x^{p_1}}{p_1!}\right)\left(1 + x + \frac{x^2}{2!} + ... + \frac{x^{p_2}}{p_2!}\right)...\left(1 + x + \frac{x^2}{2!} + ... + \frac{x^{p_k}}{p_k!}\right).$$

- **<u>The Pigeon-Hole Principle</u>**:

(1) If more than n objects are distributed into n compartments, then atleast one compartment must receive more than one objects.

(2) If $r.n+1 \quad (r \geq 1)$ pigeons are distributed among n pigeon holes, one of the pigeon holes will contain atleast $r+1$ pigeons.

(3) If n pigeons are placed in m pigeon holes, then atleast one pigeon hole will contain more than $\left[\frac{n-1}{m}\right]$ pigeons, where $[x]$ denotes the greatest integer x.

- **<u>The Principle of Inclusion and Exclusion</u>**: If $A_1, A_2, A_3, ..., A_n$ are finite sets, then

$$\left|A_1 \cup A_2 \cup A_3 \cup .... \cup A_n\right| = \sum_{1 \leq i \leq n} \left|A_i\right| - \sum_{1 \leq i \leq j \leq n} \left|A_i \cap A_j\right| + \sum_{1 \leq i \leq j \leq k \leq n} \left|A_i \cap A_j \cap A_k\right| - ....$$
$$...... + (-1)^{n+1} \left|A_1 \cap A_2 \cap A_3 \cap .... \cap A_n\right|$$

- **<u>Sieve-Formula</u>**: If $A_1, A_2, A_3, ..., A_n$ are finite subsets of se A containing N elements, then

$$\left|A_1' \cap A_2' \cap A_3' \cap .... \cap A_n'\right| = N - \sum_{1 \leq i \leq n} \left|A_i\right| + \sum_{1 \leq i \leq j \leq n} \left|A_i \cap A_j\right| - \sum_{1 \leq i \leq j \leq k \leq n} \left|A_i \cap A_j \cap A_k\right| - ...$$
$$... + (-1)^n \left|A_1 \cap A_2 \cap A_3 \cap .... \cap A_n\right|$$

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